Heat Transfer Coefficient for Laminar Flow in a Pipe

Question

In this example, we shall use pdepe to solve the governing equation for the standard boundary conditions: (i) constant wall temperature and (ii) constant heat flux. It is assumed that we have laminar flow with Re = 40 and Pr = 5, and the pipe radius R = 0.01 m and the pipe length L = 0.5 m. In addition, we assume that the boundary condition for constant heat flux is $q_w = 10\ W/m^2$, the boundary condition for constant wall temperature is $T_w = 40℃$, the entry condition is $T(\xi, 0) = 20℃$, and $k = 0.6\ W/(m\ K)$.

Solution

- **Governing Principles**

  The analysis starts with the energy equation in cylindrical coordinates, assuming steady, incompressible, laminar flow with constant properties and negligible viscous dissipation. The velocity profile for fully developed laminar flow in a pipe is parabolic:

$$u(r) = 2u_m\left(1 - \frac{r^2}{R^2}\right)$$

  where u_m is the mean velocity, R is the pipe radius, and r is the radial coordinate.

  The energy equation simplifies to:

$$u(r)\frac{\partial T}{\partial z} = \alpha\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial T}{\partial r}\right)$$

  where $\alpha = k/(\rho c_p)$ is the thermal diffusivity.

  In the **thermally fully developed region**, the shape of the dimensionless temperature profile is constant along the axial direction (z). This leads to a significant simplification for each boundary condition.

- **Case 1: Constant Wall Heat Flux**

  1. **Simplify the Energy Equation**

     Substitute the velocity profile and the constant axial gradient into the energy equation:

$$2u_m\left(1 - \frac{r^2}{R^2}\right)\frac{dT_m}{dz} = \frac{\alpha}{r}\frac{d}{dr}\left(r\frac{dT}{dr}\right)$$

  2. **Integrate to Find the Temperature Profile**

     We integrate this ordinary differential equation twice with respect to r.

First integration:

$$\int_0^r \frac{2u_m}{\alpha}\frac{dT_m}{dz}\left(r' - \frac{r'^3}{R^2}\right)dr' = \int_0^r d\left(r'\frac{dT}{dr}\right)$$

$$\frac{2u_m}{\alpha}\frac{dT_m}{dz}\left(\frac{r^2}{2} - \frac{r^4}{4R^2}\right) = r\frac{dT}{dr}$$

The boundary condition of symmetry, $\left.\frac{dT}{dr}\right|_{r=0} = 0$, sets the first integration constant to zero.

Second integration:

$$\frac{dT}{dr} = \frac{2u_m}{\alpha}\frac{dT_m}{dz}\left(\frac{r}{2} - \frac{r^3}{4R^2}\right)$$

$$T(r) - T_w = \int_R^r \frac{2u_m}{\alpha}\frac{dT_m}{dz}\left(\frac{r'}{2} - \frac{r'^3}{4R^2}\right)dr'$$

$$T(r) - T_w = \frac{2u_m}{\alpha}\frac{dT_m}{dz}\left[\frac{r^2}{4} - \frac{r^4}{16R^2}\right]_R^r$$

$$T_w - T(r) = \frac{u_m R^2}{2\alpha}\frac{dT_m}{dz}\left[\left(1 - \frac{r^4}{R^4}\right) - \frac{1}{2}\left(1 - \frac{r^2}{R^2}\right)\right]$$

3. Relate Axial Gradient to Heat Flux

An overall energy balance on a section of the pipe gives $q_w''(2\pi R) = \dot{m}c_p\frac{dT_m}{dz}$. Since $\dot{m} = \rho u_m(\pi R^2)$, we get:

$$\frac{dT_m}{dz} = \frac{2q_w''}{\rho c_p u_m R} = \frac{2\alpha q_w''}{k u_m R}$$

4. Calculate $T_w - T_m$

The bulk mean temperature T_m is defined as $T_m = \frac{\int u(r)T(r)dA}{\int u(r)dA}$. Calculating this integral with the derived temperature profile gives the difference between the wall and mean temperatures:

$$T_w - T_m = \frac{11}{24}\frac{q_w''R}{k}$$

5. Calculate Nusselt Number (N_uD)

The heat transfer coefficient is defined by $q_w'' = h(T_w - T_m)$.

$$h = \frac{q_w''}{T_w - T_m} = \frac{q_w''}{\frac{11}{24}\frac{q_w''R}{k}} = \frac{24}{11}\frac{k}{R}$$

The Nusselt number is $N_uD = \frac{hD}{k} = \frac{h(2R)}{k}$.

$$N_uD = \frac{\left(\frac{24}{11}\frac{k}{R}\right)(2R)}{k} = \frac{48}{11}$$

Analytical Result

$$N_u D = \frac{48}{11} \approx 4.364$$

- **Case 2: Constant Wall Temperature** (T_w = constant)

  This solution is more complex and is the classical Graetz problem. The solution for the fully developed region is found by solving the energy equation for the "first mode" or "first eigenvalue" of the infinite series solution. The derivation is beyond a typical hand calculation, but the result is a well-established classical value.

  1. **Problem Formulation**: The problem is solved by assuming a solution of the form $\theta(r,z) = C_n\phi_n(r)exp\left(-\lambda_n^2 \frac{\alpha z}{u_m R^2}\right)$, where $\theta = \frac{T-T_w}{T_{inlet}-T_w}$. For the fully developed region, only the first term $(n = 0)$ of the series is significant.
  2. **Solve for Temperature Profile**: The radial part of the solution involves solving a Sturm-Liouville problem to find the eigenvalues λ_n and eigenfunctions $\phi_n(r)$.
  3. **Calculate Nusselt Number**: After finding the temperature profile corresponding to the first eigenvalue, the heat transfer coefficient is calculated from the temperature gradient at the wall:

  $$h = \frac{-k\left.\frac{\partial T}{\partial r}\right|_{r=R}}{T_w - T_m}$$

  The final result, using the diameter *D* as the characteristic length, is:

Analytical Result

$$N_u D \approx 3.66$$

Result

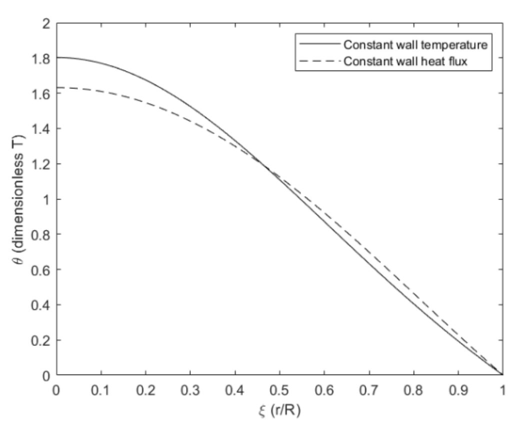

Figure 1 Temperature versus radial position for laminar pipe flow with different boundary conditions: (i) constant wall temperature and (ii) constant wall heat flux.

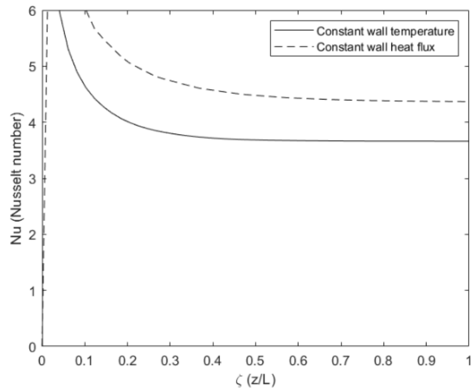

Figure 2 Nusselt number versus axial position for laminar pipe flow with different boundary conditions: (i) constant wall temperature and (ii) constant wall heat flux.